



8520 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	55604	NIRSpec MultiObject Spectroscopy	(1) 55604_MSA
	2	49410	NIRSpec MultiObject Spectroscopy	(2) 49140_MSA
	3	42046	NIRSpec MultiObject Spectroscopy	(3) 42046_MSA

ABSTRACT

JWST's discovery of a population of red compact galaxies at redshifts $z > 3$ has challenged our understanding of galaxy and AGN evolution in the early universe. This population, referred to as Little Red Dots (LRDs), consists of objects with blue rest-frame-UV slopes and red rest-frame-optical slopes, resulting in a unique spectral-energy-distribution (SED). A large fraction ($\sim 80\%$) of the LRDs show broad Balmer lines, indicating that they may host broad-line AGN that dominate their SEDs. However, some of these LRDs also show strong Balmer breaks, which instead suggest that post-starburst stellar populations dominate their SEDs. However, a recent theory has proposed that these Balmer breaks may instead be produced by clumpy dense gas near the AGN without the need for an evolved stellar population.

We propose to test this theory directly by obtaining deep (4-8 hour) NIRSpec fixed slit medium resolution spectra of the Balmer breaks of three bright LRDs (and MSA spectra of other field targets to maximize the legacy value of this program). We will robustly detect the continuum emission red of the Balmer break and search for Balmer and Ca II H and K absorption features. If Ca II lines are clearly detected, the Balmer breaks in these objects result from post-starburst stellar populations. However, if the Ca II lines are missing and the Balmer absorption lines show a velocity offset relative to strong metal emission features, the Balmer breaks result from an outflow of dense, neutral gas near the AGN. Through these observations, this program will distinguish between these competing models and directly determine the physical origins of Balmer breaks in AGN-hosting LRDs.

OBSERVING DESCRIPTION

We will obtain deep NIRSpec Fixed-Slit and MSA medium resolution grating spectra of three bright, interesting Little Red Dots (LRDs) in the EGS field (RUBIES-EGS-42046, RUBIES-EGS-49140, and RUBIES-EGS-55604). We will use these observations to resolve the continuum emission of these objects near the Ca II H and K absorption lines to determine the physical origins of the Balmer breaks exhibited by these objects. To maximize the legacy value of these observations, we will also use the MSA to observe nearby ancillary targets.

We target a S/N of 7 per pixel at rest-frame 4000Å in our primary targets so that we can robustly detect the Ca II H and K absorption lines. We can achieve this S/N with eight hour exposures for targets 42046 (using G235M/F170LP) and 49140 (using G395M/F290LP) and a four hour exposure for target 55604 (using G395M/F290LP). For the eight hour observations, we will use the standard fixed-slit+MSA three shutter nod (three exposures) with 16 groups of the NRSIRS2 readout mode and eight integrations per nod. For the four hour observation, we will use the same configuration with four integrations per nod.

Proposal 8520 - Targets - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	55604_MSA	RA: 14 19 48.7342 (214.9530592d) Dec: +52 56 12.11 (52.93670d) Equinox: J2000		
	Comments: Description=[]				
	(2)	49140_MSA	RA: 14 19 31.8652 (214.8827717d) Dec: +52 51 29.39 (52.85816d) Equinox: J2000		
	Comments: Description=[]				
	(3)	42046_MSA	RA: 14 19 10.3660 (214.7931917d) Dec: +52 46 2.85 (52.76746d) Equinox: J2000		
	Comments: Description=[]				

Proposal 8520 - Observation 1 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Observation	Proposal 8520, Observation 1: 55604										Tue Apr 21 23:00:52 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	55604_MSA	RA: 14 19 48.7342 (214.9530592d) Dec: +52 56 12.11 (52.93670d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	55604_Primary (1374 sources)		jwst-nirspec-g395m	1.5			
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	2511	214.952404	52.971475	24.83688053235422	1	6502	214.956468	52.954024	25.29519895357639	2
	1	2519	214.933776	52.958240	23.98832639179937	1	11044	214.952850	52.929603	25.13191324678153	
	1	3041	214.914463	52.941985	24.98554039961398	1	11193	214.935974	52.917004	25.16955360237213	7
	1	6189	214.956718	52.955900	25.48897480029326	1	12385	214.955307	52.924882	25.31855625600620	7
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395M/F290LP)	c1	3 Shutter Slitlet	214.95653879166 665 Degrees 52.945310277777 77 Degrees	332.25839664169 814			3	12	14180.401

Proposal 8520 - Observation 1 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Special Requirements	MSA Scheduled Aperture PA 332.2556 to 332.2556 Degrees (V3 193.68105 to 193.68105)
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Proposal 8520 - Observation 2 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Observation	Proposal 8520, Observation 2: 49410											Tue Apr 21 23:00:53 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	49140_MSA	RA: 14 19 31.8652 (214.8827717d) Dec: +52 51 29.39 (52.85816d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold				
	MSATA	false	No	MSA Center	49140_MSA (22698 sources)		jwst-nirspec-g395m	1.5				
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	8229	214.867860	52.882304	25.20422953552471	1	12540	214.830600	52.835295	24.86165822056018		
	1	8555	214.833831	52.856271	24.13726368597139	1	16016	214.865176	52.840190	23.91035661345940		
	1	9393	214.865200	52.874568	24.98000733428055	1	22785	214.896006	52.825604	23.27705529164145		
	1	11615	214.850666	52.854750	25.08849391052664	1	23413	214.898489	52.824884	25.19772437010940		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (G395M/F290LP)	c1	3 Shutter Slitlet	214.88158254166 666 Degrees 52.853688888888 89 Degrees	287.53769798087 85			3	24	28360.802	

Proposal 8520 - Observation 2 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Special Requirements	MSA Scheduled Aperture PA 287.5386 to 287.5386 Degrees (V3 148.96407 to 148.96407)
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Proposal 8520 - Observation 3 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Observation	Proposal 8520, Observation 3: 42046											Tue Apr 21 23:00:53 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	42046_MSA	RA: 14 19 10.3660 (214.7931917d) Dec: +52 46 2.85 (52.76746d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: CLEAR; Readout: NRSRAPIDD1; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD1	3	1	4	257.682		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images		Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold		
	MSATA	false	No		MSA Center	42046_Primary (1320 sources)		jwst-nirspec-g235m		1.5		
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	11913	214.758177	52.787214	24.30822561040488	1	17216	214.742119	52.745215	22.8432955262604		
	1	13005	214.729405	52.760396	24.2266963046388	1	22957	214.838119	52.783594	23.822557952550994		
	1	13947	214.765472	52.780397	21.993219857832838	1	24289	214.794482	52.746894	22.510963585303717		
	1	16283	214.779727	52.777642	22.8248661945131	1	24855	214.781468	52.735445	22.48024277513148		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time	
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	214.78266862499999 Degrees 52.76550388888889 Degrees	290.5634849383816			3	24	28360.802	

Proposal 8520 - Observation 3 - Balmer Breaks in Little Red Dots: Stellar Populations or Dense Neutral Gas?

Special Requirements	MSA Scheduled Aperture PA 290.5718 to 290.5718 Degrees (V3 151.99728 to 151.99728)
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